

LoRA: Low-Rank Adaptation of Large Language Models

Github Repo: <https://github.com/austinlu2005/myLoRA> | Austin Lu (al2622), Jason Chen (yc2786)

1. Introduction

Full fine-tuning is effective for adapting pre-trained models, but it is expensive because every parameter must be updated and stored separately for each task. LoRA addresses this by freezing the original weights and training a small low-rank update instead. For a weight matrix W_0 , LoRA learns $\Delta W = BA$, where B and A are much smaller matrices with rank r . The forward pass is $h = W_0x + (\alpha/r)BAx$. Only A and B are trained, while W_0 stays frozen.

Our project reimplemented LoRA and reproduced its main claim: low-rank adaptation can achieve performance close to full fine-tuning while training less than 1% of the parameters. We reproduced RoBERTa-base on GLUE and GPT-2 Medium on the E2E NLG challenge. We also extended LoRA to a vision-language setting by applying it to CLIP ViT-B/32 on a Pokemon image-text description dataset.

2. Chosen Result

We chose the paper's main parameter-efficiency result because it is the core motivation for LoRA. Specifically, we targeted RoBERTa results on GLUE and GPT-2 Medium results on E2E NLG. These settings test LoRA on both encoder-only and decoder-only transformer models. If LoRA can match these baselines while updating only a small number of parameters, then it provides a practical alternative to full fine-tuning for large models.

3. Methodology

We implemented LoRA from scratch by wrapping linear projections with two trainable low-rank matrices. The original weight matrix was frozen. Matrix A was initialized with Kaiming initialization, and B was initialized to zero so the model initially matched the original pre-trained model. We used rank $r = 8$, $\alpha = 16$, dropout 0.1, AdamW, and a linear warmup-decay schedule across experiments.

For RoBERTa-base, LoRA was inserted into the attention query and value projections and evaluated on GLUE tasks such as MNLI, SST-2, MRPC, CoLA, QNLI, QQP, RTE, and STS-B. For GPT-2 Medium, LoRA was inserted into the `c_attn` projection, GPT-2's fused query-key-value attention layer. This slightly differs from the paper's query/value-only setup, but it matched the HuggingFace GPT-2 implementation more naturally.

For the extension, we applied LoRA to CLIP ViT-B/32, which contains a Vision Transformer image encoder and a text encoder. We inserted LoRA into the query and value projections of both encoders and trained on a Pokemon image-text description dataset. Each Pokemon image was paired with a short text description, and the model was trained using contrastive learning: correct image-text pairs were pulled closer together, while incorrect pairs were pushed apart. We evaluated CLIP using Recall@1 and Recall@5 for both image-to-text and text-to-image retrieval.

4. Results and Analysis

RoBERTa on GLUE closely matched the LoRA paper on most tasks. MNLI was 87.9 compared to the paper's 87.5, SST-2 was 94.8 compared to 95.1, QNLI was 92.8 compared to 93.3, QQP was 90.8 compared to 90.8, and STS-B was 90.9 compared to 91.5. The main gaps were MRPC, 88.2 compared to 89.7, and RTE, 79.1 compared to 86.6. We believe these gaps mainly come from limited reproduction resources: our results were single-seed, while the original paper reports more stable multi-seed results. RTE and MRPC are also smaller datasets, making them more sensitive to seed and learning-rate choices. GPT-2 Medium on E2E NLG also reproduced the original results closely. BLEU was 69.78 compared to 70.40, NIST was 8.79 compared to 8.85, METEOR was 45.62 compared to 46.00, ROUGE-L was 71.29 compared to 71.80, and CIDEr was 2.48 compared to 2.53. These small differences suggest that our LoRA implementation successfully reproduced the generation result, even though our GPT-2 injection adapted the fused attention projection rather than only separate query and value matrices.

The CLIP extension showed that LoRA can also help Vision Transformer-based vision-language models. On the Pokemon image-text description dataset, image-to-text Recall@1 improved from 0.1750 to 0.5562, and text-to-image Recall@1 improved from 0.4250 to 0.5312. Mean Recall@1 increased from 0.3000 to 0.5437. However, this was an extension rather than a direct reproduction because the original LoRA paper did not evaluate CLIP. Since we compared against zero-shot CLIP but not full fine-tuning, the result shows that LoRA improves this CLIP model on the dataset, but not necessarily that it matches full fine-tuning.

Overall, the parameter-efficiency claim was reproduced. RoBERTa used 294,912 trainable parameters, about 0.24% of the model; GPT-2 Medium used 786,432, about 0.22%; and CLIP ViT-B/32 used 491,520, about 0.33%. Across all three settings, LoRA reduced trainable parameters by about 99.7% while maintaining strong performance.

5. Reflections

This project showed that LoRA is simple mathematically but careful to reproduce in practice. The core update $\Delta W = BA$ is short, but final performance depends on implementation details such as which layers are adapted, how the matrices are initialized, whether the task head is trained, and how learning rates are selected. Our largest discrepancy, RTE, also showed that small datasets can have high variance, so a faithful reproduction should use multiple seeds and report medians.

We also learned that extending LoRA to new architectures requires practical choices. GPT-2's fused attention layer made exact query/value-only adaptation harder, and CLIP required deciding whether to adapt the vision encoder, text encoder, or both. Applying LoRA to both CLIP encoders gave clear retrieval gains on the Pokemon dataset, suggesting that low-rank adaptation can transfer beyond language models to multimodal Vision Transformer settings.

With more time, we would run five seeds for GLUE, test different ranks such as $r = 1, 2, 4, 8$, and 16, add a full fine-tuning baseline for CLIP, and implement a structured GPT-2 LoRA module that updates only the query and value slices. Overall, our reimplementation supports the paper's main claim: LoRA can achieve strong downstream performance while training less than 1% of the parameters, and it may also be useful for efficient adaptation of vision-language models.

6. References

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